Preface

When two of your authors, Neal and Mark, were writing the book *Fundamentals of Software Architecture*, we kept coming across complex examples in architecture that we wanted to cover but that were too difficult. Each one offered no easy solutions but rather a collection of messy trade-offs. We set those examples aside into a pile we called "The Hard Parts." Once that book was finished, we looked at the now gigantic pile of hard parts and tried to figure out: *why are these problems so difficult to solve in modern architectures*?

We took all the examples and worked through them like architects, applying trade-off analysis for each situation, but also paying attention to the process we used to arrive at the trade-offs. One of our early revelations was the increasing importance of data in architecture decisions: who can/should access data, who can/should write to it, and how to manage the separation of analytical and operational data. To that end, we asked experts in those fields to join us, which allows this book to fully incorporate decision making from both angles: architecture to data and data to architecture.

The result is this book: a collection of difficult problems in modern software architecture, the trade-offs that make the decisions hard, and ultimately an illustrated guide to show you how to apply the same trade-off analysis to your own unique problems.

Conventions Used in This Book

The following typographical conventions are used in this book:

Italic

Indicates new terms, URLs, email addresses, filenames, and file paths.

`Constant width`

Used for program listings, as well as within paragraphs to refer to program elements such as variable or function names, databases, data types, environment variables, statements, and keywords.

Constant width bold

Shows commands or other text that should be typed literally by the user.

Constant width italic

Shows text that should be replaced with user-supplied values or by values determined by context.

TIP

This element signifies a tip or suggestion.

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